



NYC Yellow Taxi – Big Data Analytics

BITS ZG522 · Big Data Systems · Assignment 1 · 3 months (Jan–Mar 2026) · ~11M trips

Total Trips

7,799,600

Revenue (USD)

\$226.64 M

Avg Fare

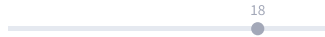
\$40.71

Avg Tip %

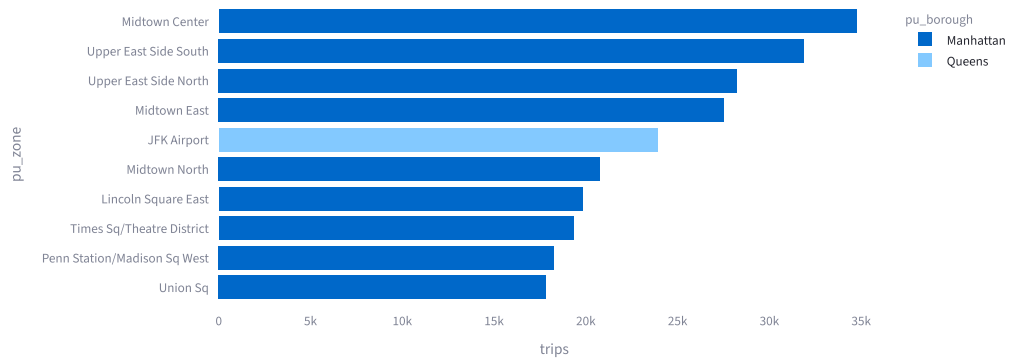
4.5%

Q1 · Demand hotspots by hour

Pickup hour

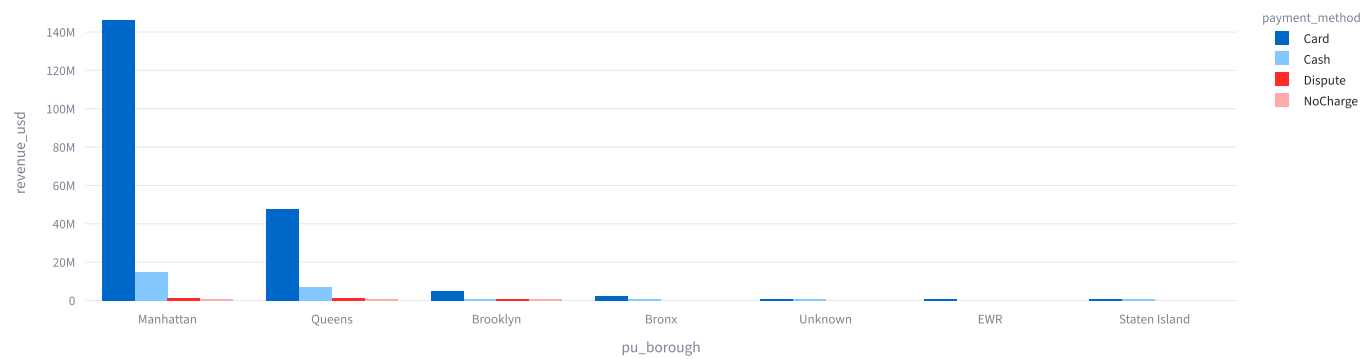


Top 10 pickup zones at 18:00



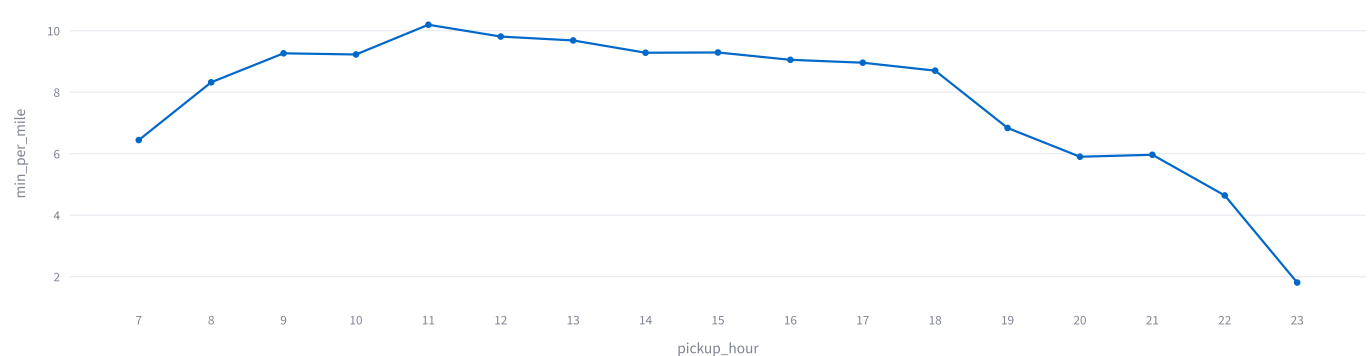
Q2 · Revenue by borough and payment method

Revenue (USD) by borough × payment



Q3 · Congestion – minutes per mile by hour

Avg minutes per mile across all busy PU→DO pairs



Top 15 slowest PU→DO pairs (min per mile)

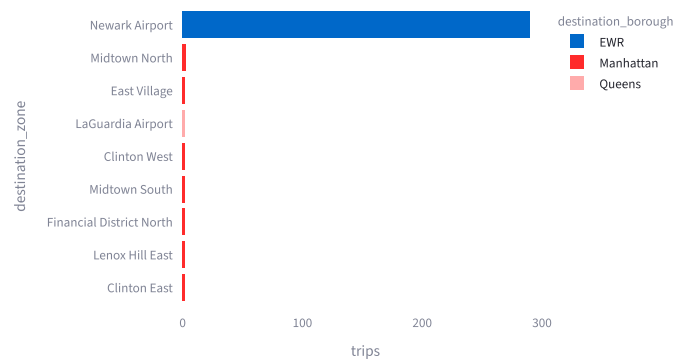
	pickup_hour	pu_loc_id	do_loc_id	trips	avg_duration_min	avg_distance_mi	min_per_mile
100	12	186	230	1738	16.66	1	18.07
354	17	161	161	1179	8.5	0.55	17.95
91	13	186	230	1772	16.12	1	17.75
116	11	186	230	1685	16.22	1	17.75
169	18	161	230	1497	11.42	0.73	17.34
465	13	161	161	1051	8.44	0.56	17.2
359	16	161	161	1174	8.04	0.58	16.46
404	18	162	230	1124	13.4	0.87	16.37
247	11	186	161	1325	19.34	1.24	16.33
462	18	161	161	1055	7.74	0.58	15.73

Q4 · Airport trip profile

Airport

☐ JFK ☐ LGA ☒ EWR

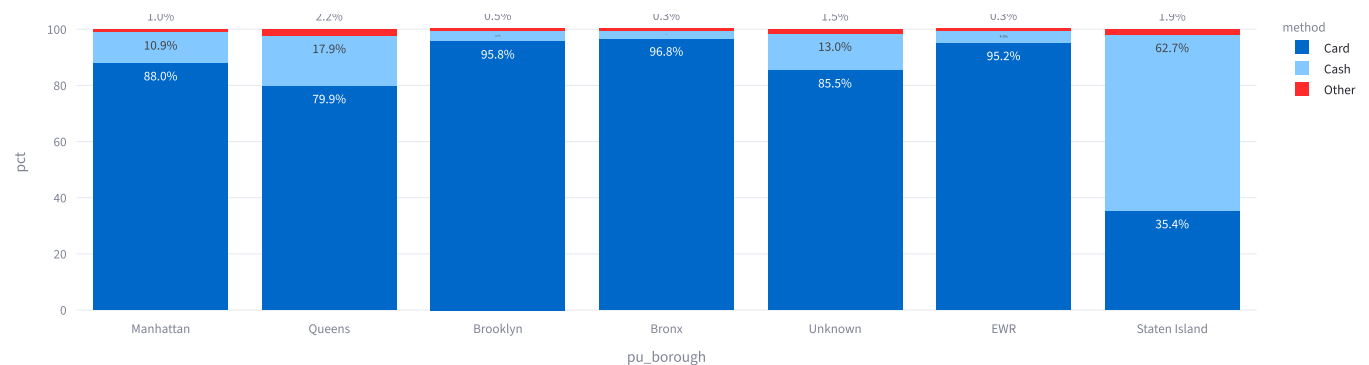
Top destinations from EWR



	airport	destination_zone	destination_borough	trips	avg_distance_mi
0	EWR	Newark Airport	EWR	290	0.27
1	EWR	Outside of NYC	None	25	6.49
2	EWR	Midtown North	Manhattan	3	17
3	EWR	East Village	Manhattan	2	15.77
12	EWR	Clinton East	Manhattan	1	17.82
18	EWR	LaGuardia Airport	Queens	1	26.02
17	EWR	Lenox Hill East	Manhattan	1	21.38
16	EWR	Financial District North	Manhattan	1	13.57
15	EWR	Midtown South	Manhattan	1	16.6
14	EWR	Clinton West	Manhattan	1	17.27

Q5 · Card vs cash share by borough

Payment method share (%)



Pipeline: HDFS → Pig (ETL) → native MR (trips-per-zone) → Hive (analytics) → HBase (zone lookup) → Streamlit